
DAWN-SRW: A Computational Model of Learned State Transitions

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Abstract

Dense language models operate over learned high-dimensional states, but the operations producing their transitions and the conditions governing their use remain largely implicit. This motivates a broader computational question: *Can computation over learned high-dimensional states be organized as an explicit, learnable system of state-conditioned control over operations?* DAWN-SRW tests this question in language modeling through a narrower architectural hypothesis: transition-producing language-model computation can be factorized into identifiable learned operations and explicit state-conditioned control that determines how those operations contribute to the subsequent state transition. Here, identifiable means referable as a stable architectural object and available for measurement or intervention, not necessarily human-interpretable. The evaluated v4172 model is one concrete realization. It uses persistent rank-1 read-write (RW) banks, learned operation-address control, continuous-valued applicability coefficients with exact support, and normalized additive composition to construct Q/K/V attention interfaces and direct RST updates; operator visibility remains global and RW execution remains dense. At 393.8M parameters and 39.999B tokens, DAWN-SRW retains performance against a matched dense Transformer: C4 cross-entropies are 2.9402 versus 2.9359, and six-task zero-shot means are 0.4629 versus 0.4597. A descriptive 1.272B evaluation reaches 2.8359 cross-entropy and a 0.4858 zero-shot mean. The model exhibits exact input-conditioned applicability support, and a discovery-frozen 4,096-site Q/K-centered operator circuit passes suppression, four matched-control families, and exact restoration on independent validation and held-out indirect-object-identification inputs. Fail-closed results delimit where localization did not succeed. Together, these results provide a constructive test of the

narrower architectural hypothesis under the evaluated conditions and evidence relevant to the broader computational question, without establishing human-readable semantics, general editability, benchmark superiority, or hardware efficiency.

1. Introduction

Modern language models achieve strong performance by repeatedly transforming learned high-dimensional representations, but those transformations do not expose the constituent operations or the conditions under which a state uses them. The long-term motivation is to make learned computation available as explicit computational objects that can be identified, analyzed, and causally intervened on. Such an interface could eventually support semantic interpretation, reliable editing, or more programmable learned systems. Those outcomes are motivations, not claims established by this paper.

We begin from a broader computational question:

Can computation over learned high-dimensional states be organized as an explicit, learnable system of state-conditioned control over operations?

At this level, a learned representation x is the current computational state, the state determines an applicability profile $C(x) = (c_1(x), \dots, c_N(x))$ that represents state-conditioned control, and controlled operation contributions construct the computation that produces the subsequent state transition:

$$x \longrightarrow C(x) \longrightarrow \text{controlled operations} \longrightarrow x'.$$

Control answers, at the architectural level, which learned operations are applicable to the current state and to what degree; the operations determine what computation is performed on that state. DAWN-SRW studies this broad question in the concrete domain of language modeling.

At a high level, conventional programming makes computation explicit through states, operations, and rules governing their execution. Operational accounts describe execution

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through transitions between configurations (Plotkin, 1981):

current state $\longrightarrow$ conditions / applicable operations
 $\longrightarrow$ execution $\longrightarrow$ next state.

Such explicit structure supplies objects through which computation can be referred to and analyzed. We use this program-like organization only in the limited sense shown above—state, conditions and applicable operations, execution, and next state. It does not imply symbolic programs, discrete or human-readable instructions or predicates, or a formal operational semantics for DAWN-SRW.

Language models likewise evolve computational states by repeatedly transforming representations across depth. Let X_ℓ denote the sequence representation at layer ℓ ; at a coarse level, a block implements

$$X_\ell \longrightarrow X_{\ell+1}.$$

We likewise use this state-evolution view only as an organizing observation, not as a contribution or a claim about an intrinsic language-model semantics.

In conventional Transformers, the mechanisms producing these state changes remain largely implicit in dense parameterized transformations. Attention and feed-forward layers expose module boundaries, but they do not natively expose a referable set of operations together with explicit state-conditioned control over their application.

We therefore study a narrower architectural hypothesis:

Can transition-producing language-model computation be factorized into identifiable learned operations and explicit state-conditioned control that determines how those operations contribute to the subsequent state transition?

This hypothesis separates what computation is available from when and how it is applied. “Identifiable” means that an operation is referable as a stable architectural object whose contribution can be measured or intervened on; it does not specify how broadly or how long that identity must persist, and does not imply a human-readable concept, a discrete instruction, or a uniquely correct decomposition. The paper provides a constructive architectural test, not evidence that this factorization is intrinsic to all language models.

The evaluated DAWN-SRW v4172 architecture instantiates one point in this broader design space. It chooses a rank-1 read-write transformation as its RW primitive and normalized additive composition as its application semantics. For

a token state x ,

$$\mathcal{O}_i(x) = \langle x, r_i \rangle w_i, \quad (1)$$

$$F(x) = \frac{1}{Z(x)} \sum_i c_i(x) \mathcal{O}_i(x), \quad c_i(x) \in [0, 1]. \quad (2)$$

The read direction r_i determines what is extracted from the current representation, the write direction w_i determines the direction of the primitive’s output contribution, and $c_i(x)$ determines whether and to what degree that primitive applies. Thus, this realization parameterizes transition-producing computation through explicit conditional composition of RW primitives.

The control abstraction is also explicit:

$$x \mapsto C_a(x) = (c_{a,1}(x), \dots, c_{a,N_a}(x)). \quad (3)$$

These components instantiate the narrower hypothesis: the representation supplies the current state, $C_a(x)$ controls primitive application, and the composed result supplies computation used in the next update. Continuous-valued coefficients with exact support and normalized addition are v4172 design choices, not requirements of the architectural hypothesis.

More specifically, v4172 uses persistent type-specific QK, V, and RST RW banks across depth; computation-type-specific projected state queries; read/write-derived primitive addresses in a learned spherical geometry; continuous state-dependent thresholds; exact applicability support with continuous-valued coefficients; normalized additive composition; and deployment across Q, K, V, and the direct residual-state transition (RST). The evaluated realization does not independently predict an unconstrained applicability coefficient for each primitive. Instead, it parameterizes applicability through a shared learned geometric relation between a projected state query and the primitive addresses. Comparing that continuous relation with the state-dependent threshold creates exact support membership, while normalized positive margin determines continuous-valued applicability within the support. The architecture thereby specifies the form through which applicability is expressed; the resulting state–operation organization is learned from the language-model objective rather than assigned semantic conditions a priori. The operation-address space implements control rather than RW execution. The current model evaluates applicability with global operator visibility and applies the RW banks densely, even when exact applicability support is a strict subset of the pool.

This realization builds on the read-write view of dense feed-forward computation, where first-layer directions detect input patterns and second-layer directions contribute outputs (Geva et al., 2021). That analysis reveals functional directions inside an existing dense layer. DAWN-SRW instead turns RW factorization into an architectural constraint:

reusable RW units are executable RW primitives whose state-dependent contributions are directly controlled and observable.

The same formulation produces the query, key, and value states used by causal self-attention and the direct update applied to the post-attention residual state. Q and K produce separate outputs from one shared QK pool; V and the post-attention residual transition (RST) use separate pools. Operator pools and applicability projections are reused across depth, while normalization and attention output projections remain layer-specific.

The empirical study follows a fixed claim hierarchy. First, viability is tested by comparing a 393.8M parameter DAWN-SRW checkpoint with a parameter- and token-matched dense Transformer on C4 and six zero-shot tasks, with a descriptive 1.272B evaluation. Second, conditional organization is tested by measuring exact input-conditioned applicability support. Third, causal accessibility is tested by localizing a discovery-only indirect object identification (IOI) circuit. The circuit is frozen before confirmatory data are opened, then tested with suppression, matched controls, and exact restoration on independent validation and fully held-out test inputs. Fourth, three fail-closed boundaries are reported: RAVEL localization, ARC causal validation, and 1.3B IOI localization. Efficiency is not a headline result: the current implementation scores and applies full operator pools densely.

Our contributions are:

1. **An explicit architectural factorization and one realization.** We formulate transition-producing computation in terms of identifiable learned operations and state-conditioned control that determines how those operations contribute to the subsequent state transition. We instantiate that factorization in v4172 through persistent rank-1 RW banks, learned spherical operation-address control, exact applicability support with continuous-valued coefficients, and normalized composition across Q/K/V/RST.
2. **A constructive empirical test with bounded evidence.** We test viability at 400M scale, report a descriptive 1.3B evaluation, characterize conditional organization through applicability support, and test causal accessibility through a discovery-frozen IOI circuit confirmed by suppression, matched controls, and exact restoration. Fail-closed results bound where the same localization program did not succeed.

2. Related Work

2.1. State-dependent and dynamically parameterized transformations

State-dependent parameterizations have long precedents in nonlinear control and neural sequence models. State-dependent Riccati equation techniques use state-dependent coefficient representations in nonlinear control design (Cloutier, 1997); we cite this as conceptual lineage, not as a mathematical equivalence to DAWN-SRW. Multiplicative RNNs allow the current input to determine the hidden-state transition matrix (Sutskever et al., 2011), input-switched affine networks use input-dependent recurrent weights (Förster et al., 2017), and HyperNetworks generate weights for another network (Ha et al., 2017). These works address different settings—feedback control, recurrent dynamics, and generated neural parameters—but collectively establish that a transformation may depend on the current state or input. State evolution, state dependence, and dynamically generated transformations are therefore not contributions of DAWN-SRW. The narrower hypothesis studied here is that transition-producing language-model computation can be factorized into identifiable learned operations and explicit state-conditioned control that determines how those operations contribute to the subsequent state transition. Persistent RW primitives, address-based control, and normalized additive composition are choices of the evaluated realization rather than the hypothesis itself.

2.2. Conditional computation and fine-grained experts

Sparsely-gated mixtures of experts activate a routed subset of expert networks (Shazeer et al., 2017), and Switch Transformers scale this pattern with simple routing (Fedus et al., 2022). PEER applies product-key retrieval to more than a million tiny experts (He, 2024); OmniMoE extends this fine-grained line with vector-level Atomic Experts, a Cartesian Product Router, and execution scheduling (Shi et al., 2026). Thus state-dependent routing, fine-grained experts, large expert pools, and sparse retrieval all precede DAWN-SRW. These systems primarily develop routing, capacity, and efficient sparse execution for expertized layers. The paper instead studies explicit state-conditioned operator organization as the base parameterization of transition-producing computation, and exposes the resulting operator contributions for causal intervention. Its current implementation still performs a dense address scan and dense, chunked RW application rather than indexed sparse execution.

2.3. Shared computation across depth

Parameter sharing across depth and layer-agnostic expert reuse also have direct precedents. Universal Transformers recur over shared layer parameters (Dehghani et al., 2019),

while MoEUT develops a shared-layer Transformer with expertized feed-forward and attention computation (Csordás et al., 2024a). More recent MoE designs move expert capacity beyond layer-local ownership: MoUE reuses a universal layer-agnostic expert pool across layers (Chen et al., 2026), and UniPool exposes one global pool through independent per-layer routers (Huang et al., 2026). The former line studies recurrence or shared-layer computation, while the latter relaxes layer-local expert ownership. DAWN-SRW therefore does not claim depth sharing or a global pool as new. It persistently reuses fine-grained, type-specific QK, V, and RST RW banks; each depth’s current representation recomputes applicability over the same bank, and the shared operation capacity constructs Q/K/V attention interfaces or a direct residual transition.

2.4. Conditional attention and unified expertized transformations

Conditional computation has also been applied beyond Transformer feed-forward layers. SwitchHead routes among attention-side experts and reduces the number of attention matrices evaluated (Csordás et al., 2024b). UMoE reformulates attention to expose an FFN-like expert structure, then studies a shared expert set across attention and FFN computation (Yang et al., 2025). They establish that expertized transformations and parameter sharing need not be confined to the FFN. These works rule out positioning attention-side experts or conditional attention itself as a DAWN-SRW contribution. DAWN-SRW instead applies one state-conditioned RW operator-composition abstraction consistently to the learned Q, K, and V interfaces and the direct post-attention residual-state transition, with persistent type-specific RW banks reused across depth.

2.5. Read-write primitives and memory views

Transformer feed-forward layers admit a key-value-memory interpretation in which read-side directions detect input patterns and write-side directions contribute outputs (Geva et al., 2021). Product Key Memory factorizes retrieval keys (Lample et al., 2019), and scalable memory layers extend large content-addressable trainable memories (Berges et al., 2024). Fast Weight Programmers provide an algebraic and programming precedent based on additive outer-product updates (Schlag et al., 2021). MoRAM explicitly treats rank-1 adapters as associative-memory experts and uses their intrinsic keys for self-activation (Lu et al., 2026). Together these lines establish separate precedents for read/write interpretation, scalable retrieval, outer-product algebra, and rank-1 memory experts. RW, rank-1, and memory views are therefore not new individually. DAWN-SRW uses RW factorization as an executable architectural primitive rather than only a post-hoc interpretation: the architecture directly exposes each primitive’s state-dependent applicability, exe-

cuted response, and contribution to the composed computation.

2.6. Neural programming and programming interpretations

Neural Programmer-Interpreters represent and execute programs from supervised traces (Reed & de Freitas, 2016), while Neural Programmer differentially selects operations and data (Neelakantan et al., 2016). Fast Weight Programmers connect outer-product computation to elementary instructions (Schlag et al., 2021). DAWN-SRW does not induce a symbolic program, define a formal operational semantics, or require individual operators to be semantically interpretable. Here “program-like” refers only to explicit learned operations, state-conditioned applicability, and their controlled application in producing state transitions. Plotkin’s operational perspective is an organizational reference rather than a formal semantics for the model.

2.7. Mechanistic analysis and causal intervention

Mechanistic interpretability studies behavior through localized components and causal interventions. IOI work isolates an attention-head circuit and evaluates faithfulness, completeness, and minimality (Wang et al., 2023). AtP* provides a scalable approximation to activation patching for localizing behavior to model components (Kramár et al., 2024). RAVEL shows why neuron-level representations may remain entangled and evaluates methods against multiple causal criteria (Huang et al., 2024). This literature distinguishes a candidate localization from evidence that an intervention causally controls behavior. DAWN-SRW supplies precise RW-operator intervention targets, but explicit units do not guarantee semantic interpretability and localization alone is not causal confirmation. The present evidence therefore preserves a discovery freeze followed by suppression, matched controls, and exact same-example restoration, and does not generalize the IOI result beyond the specified task, model, and intervention.

Across these lines, state dependence, conditional routing, depth sharing, attention-side experts, rank-1 RW structure, and programming interpretations all have clear precedents. The contribution here is a constructive architectural test of whether such an organization can be explicitly instantiated in language modeling. The narrower tested factorization is instantiated consistently across Q/K/V/RST and evaluated for viability, conditional organization, and causal accessibility.

3. DAWN-SRW

3.1. State Evolution as the Starting Abstraction

Let

$$X_\ell = (x_{\ell,1}, \dots, x_{\ell,T}) \in \mathbb{R}^{T \times D}$$

denote the sequence representation at depth ℓ . At a coarse level, a Transformer-like block evolves this state through a relation-dependent attention update followed by a transformation of the contextualized representation:

$$Y_\ell = X_\ell + A_\ell(X_\ell), \quad (4)$$

$$X_{\ell+1} = Y_\ell + R_\ell(Y_\ell). \quad (5)$$

This decomposition only fixes the computational abstraction and notation. DAWN-SRW concerns how the transformations producing these updates are parameterized.

3.2. Explicit State-Conditioned Operator Parameterization

We use *operation* for the abstract computational object, *RW primitive* for its rank-1 realization in v4172, and *operator* when referring to an indexed primitive as an architectural object.

The architectural hypothesis is broader than any particular primitive, control geometry, or application rule. It separates identifiable learned operations from explicit state-conditioned control that determines how those operations contribute to the subsequent state transition. The evaluated v4172 architecture instantiates that hypothesis through reusable RW primitives, an applicability profile, and normalized additive composition. For a token state $x \in \mathcal{S} = \mathbb{R}^D$, let $a \in \{Q, K, V, \text{RST}\}$ index the computation type and define

$$F_a : \mathcal{S} \rightarrow \mathbb{R}^D. \quad (6)$$

The Q, K, and V outputs form the state-conditioned interfaces through which relational interaction is computed, whereas F_{RST} directly specifies a residual-state update. Each composed output is explicitly decomposed into contributions from reusable RW primitives under state-dependent control. This RW factorization, its rank-1 granularity, the operation-address coupling, spherical geometry, and normalized additive composition are design choices of v4172. They provide one concrete test of the hypothesis rather than defining it.

At the computational level, this view separates two questions for a state x : which learned operations are applicable, and what transformations those operations perform. DAWN-SRW makes both sides architecturally explicit: $C_a(x)$ represents state-conditioned control over primitive application, whereas the RW primitives represent the transformations whose contributions are controlled. Spherical operation-address geometry realizes this applicability profile in v4172;

it does not define the abstract control representation or the broader hypothesis.

3.3. Read-Write Computational Primitives

Each primitive owns learned read and write parameters. With $\text{unit}(z) = z / \max(\|z\|_2, \epsilon)$, define normalized directions $\bar{r}_{a,i} = \text{unit}(r_{a,i})$ and $\bar{w}_{a,i} = \text{unit}(w_{a,i})$. The primitive executes

$$\mathcal{O}_{a,i}(x) = \langle x, \bar{r}_{a,i} \rangle \bar{w}_{a,i} = \bar{w}_{a,i} \bar{r}_{a,i}^\top x. \quad (7)$$

This is an explicit rank-1 RW primitive. It reads a scalar from the current state and produces a directed output contribution. The read direction specifies the extracted scalar, while the write direction specifies the direction of that contribution. For Q, K, and V, these contributions compose the interfaces used by attention; for RST, they compose a direct residual-state update. Unlike a static retrieved value, an RW primitive remains an input-dependent transformation, with its output varying with the current state through the read response. The state-to-state transition is completed only when the composed output participates in the attention-mediated or residual update.

3.4. State-Conditioned Applicability

At the abstract level, control is represented by an applicability profile over reusable RW primitives. For each computation type $a \in \{Q, K, V, \text{RST}\}$, the current state induces a profile over all N_a primitives,

$$C_a(x) = (c_{a,1}(x), \dots, c_{a,N_a}(x)), \quad (8)$$

whose coordinates specify which primitives participate and to what degree, rather than assigning the state to a single discrete branch. The evaluated v4172 profile is not produced by independently predicting an unconstrained N_a -dimensional vector of per-primitive coefficients. Instead, the current state produces a computation-type-specific projected query $q_a(x)$, each primitive has a live read/write-derived address $k_{a,i}$, and the same learned geometric rule structures every coordinate of the profile:

$$q_a(x), k_{a,i} \longrightarrow \rho_{a,i}(x) \longrightarrow \mathcal{I}_a(x) \longrightarrow c_{a,i}(x).$$

The directional similarity $\rho_{a,i}(x) = q_a(x)^\top k_{a,i}$ is a continuous state–operation relation. Its comparison with a state-dependent boundary gives exact applicability-support membership, and normalized positive margin gives a continuous-valued coefficient within that support. The architecture specifies this shared form of applicability, while the resulting state–operation organization is learned from the language-model objective rather than assigned semantic conditions a priori. This is not an algebraic decomposition of the representation vector; the profile coordinates control the contributions of the corresponding RW primitives composed below.

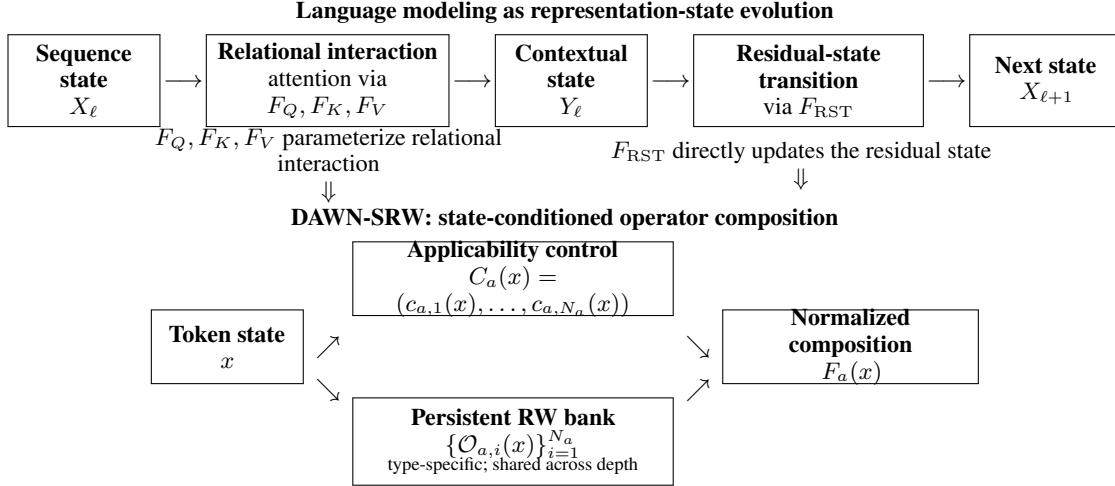


Figure 1. The two-level construction. Top: a language-model block evolves a sequence representation through relational interaction and a subsequent residual-state transition. Bottom: x separately determines the applicability profile and input-dependent responses of a persistent, type-specific RW bank shared across depth; the two meet in normalized composition $F_a(x)$. In v4172, $q_a(x)$, $\tau_a(x)$, $k_{a,i}$ implement applicability in address space; RW execution remains in the full D -dimensional representation space.

Section 3.6 gives the query and address construction. In v4172, the implementation’s `raw_tau` is a direct linear projection $t_a(x)$, which a sigmoid maps to the state-dependent DirectTau boundary in cosine space:

$$t_a(x) = xu_a + \beta_a, \quad (9)$$

$$\tau_a(x) = -1 + 2\sigma(t_a(x)), \quad (10)$$

$$\rho_{a,i}(x) = q_a(x)^\top k_{a,i}, \quad (11)$$

$$m_{a,i}(x) = \rho_{a,i}(x) - \tau_a(x). \quad (12)$$

Here $\tau_a(x)$ is a continuous state-dependent applicability-support boundary parameter, and $m_{a,i}(x)$ is the applicability margin. The evaluated `linear_angular` rule defines primitive applicability as

$$c_{a,i}(x) = \text{clip}\left(\frac{m_{a,i}(x)}{\max(1 - \tau_a(x), 10^{-4})}, 0, 1\right). \quad (13)$$

We call $c_{a,i}(x) \in [0, 1]$ the *continuous-valued applicability coefficient* for primitive i at state x . Its exact state-dependent applicability support is

$$\begin{aligned} \mathcal{I}_a(x) &= \{i : \rho_{a,i}(x) > \tau_a(x)\} \\ &= \{i : m_{a,i}(x) > 0\} = \{i : c_{a,i}(x) > 0\}. \end{aligned} \quad (14)$$

The exact property is this derived support membership, not the boundary parameter itself. At and outside the boundary, a primitive contributes exactly zero; inside it, $0 < c_{a,i}(x) \leq 1$, increasing continuously with normalized positive margin. This combines state-dependent sparse applicability support with a continuous-valued applicability coefficient rather than merely assigning a dense soft mixture over every primitive.

3.5. State-Conditioned Operator Composition

Applicable primitives compose into the output $F_a(x)$:

$$Z_a(x) = \left[\max\left(\sum_i c_{a,i}(x), 1\right) \right]^{p_a}, \quad (15)$$

$$s_a = \sqrt{D/L},$$

$$F_a(x) = \frac{s_a}{Z_a(x)} \sum_i c_{a,i}(x) \mathcal{O}_{a,i}(x). \quad (16)$$

Thus the applicability coefficient multiplies the current-state RW read response $\langle x, \bar{r}_{a,i} \rangle$, and the resulting write contributions are summed under the normalization above. Equivalently,

$$\begin{aligned} F_a(x) &= A_a(x)x, \\ A_a(x) &= \frac{s_a}{Z_a(x)} \sum_i c_{a,i}(x) \bar{w}_{a,i} \bar{r}_{a,i}^\top. \end{aligned} \quad (17)$$

This form makes the architectural hypothesis explicit. The model does not apply one fixed dense transformation to every state. Instead, the current representation determines continuous-valued applicability coefficients, which construct a state-dependent transformation $A_a(x)$ from reusable rank-1 RW primitives. The resulting computation is both input-conditioned and explicitly decomposed into identifiable operator contributions.

“Continuous-valued” refers to the applicability coefficient, not to continuous-time dynamics or a continuous-depth model; layer evolution remains discrete. The scalar mapping from normalized applicability margin to applicability is continuous and piecewise linear, with nondifferentiable support boundaries.

The denominator powers are $p_Q = p_K = 0.5$, $p_V = 1.0$, and $p_{\text{RST}} = 1.2$. The fixed composition scale is $s_a = \sqrt{D/L}$, with no learned pool-strength parameter.

3.6. Operation-Address Space

The full learned representation state remains $x \in \mathcal{S} = \mathbb{R}^D$. The operation-address space $\mathcal{A} = \mathbb{R}^R$ contains the projected query and primitive addresses used to evaluate applicability; it is not where RW computation itself is executed. This addressing construction is one evaluated realization of the state-conditioned applicability abstraction above. The present experiments do not establish that this particular realization is necessary or preferable to alternative state-conditioned control mechanisms. In v4172, $R = 256$, and a normalized query is a direct projection of the current input state:

$$q_a(x) = \text{unit}(xP_a + b_a) \in \mathcal{A}. \quad (18)$$

A shared probe pair $P_r, P_w \in \mathbb{R}^{D \times R}$ maps all operator pools' functional read and write directions into address features. The operator address is

$$\begin{aligned} u_{a,i} &= \text{unit}(\bar{r}_{a,i}^\top P_r), & v_{a,i} &= \text{unit}(\bar{w}_{a,i}^\top P_w), \\ k_{a,i} &= \text{unit}(u_{a,i} \odot v_{a,i}) \in \mathcal{A}. \end{aligned} \quad (19)$$

This generalized coordinate-wise bilinear mapping makes address identity a live learned function of what a primitive reads and where it writes, rather than an independent expert ID. It is the addressing realization used to compute the applicability coefficients in Equation 13, not the paper's definition of a state transition. Shared probes align QK, V, and RST addresses to one coordinate convention, while their operator rows remain in QK-, V-, and RST-specific pools.

Because both $q_a(x)$ and $k_{a,i}$ are unit-normalized, they lie on the address sphere $\mathbb{S}^{R-1}$. Let $\theta_{a,i}(x) = \arccos(q_a(x)^\top k_{a,i})$ and $\theta_a(x) = \arccos \tau_a(x)$. The applicability support is therefore

$$\mathcal{I}_a(x) = \{i : \theta_{a,i}(x) < \theta_a(x)\},$$

a spherical cap centered at the query $q_a(x)$ with state-dependent angular radius $\theta_a(x)$. Geometrically, $q_a(x)$ specifies the center of the applicability query, $\tau_a(x)$ specifies its applicability-support boundary, and $c_{a,i}(x)$ specifies the continuous-valued coefficient with which primitive i applies. The cap is therefore the geometric representation of exact applicability support within the globally visible pool, not a visibility boundary. When the numerical guard in Equation 13 is inactive, an applicable primitive has

$$c_{a,i}(x) = \frac{\cos \theta_{a,i}(x) - \tau_a(x)}{1 - \tau_a(x)},$$

which rises continuously from zero at the cap boundary to one at its center. Equivalently, $c_{a,i}(x)$ is a normalized

continuous depth within the cap. Equation 13 is the exact implementation, retaining the 10^{-4} denominator guard and clipping.

3.7. Attention-Mediated and Residual Transitions

For a sequence state X_ℓ , layer-specific normalization gives $\tilde{X}_\ell = \text{LN}_\ell^{(1)}(X_\ell)$. Three state-conditioned operator compositions produce the attention interfaces:

$$Q_\ell = F_Q(\tilde{X}_\ell),$$

$$K_\ell = F_K(\tilde{X}_\ell),$$

$$V_\ell = F_V(\tilde{X}_\ell),$$

$$A_\ell = \text{softmax}\left(\frac{Q_\ell K_\ell^\top}{\sqrt{d_h}} + M_{\text{causal}}\right),$$

$$\Delta X_{\text{attn},\ell} = A_\ell V_\ell W_\ell^O, \quad (20)$$

$$Y_\ell = X_\ell + \Delta X_{\text{attn},\ell}. \quad (21)$$

Q and K produce separate outputs from the same 4,600-row QK pool; V produces an output from a separate 14,500-row pool. After attention, a second layer-specific normalization $\tilde{Y}_\ell = \text{LN}_\ell^{(2)}(Y_\ell)$ is evaluated anew using the 29,300-row RST operator pool:

$$X_{\ell+1} = Y_\ell + F_{\text{RST}}(\tilde{Y}_\ell). \quad (22)$$

The block thus applies an attention-mediated state transition followed by a direct residual-state transition. RST applicability depends on the contextualized post-attention state, not the pre-attention query.

3.8. Cross-Depth Operator Reuse

The QK, V, and RST read/write pools; the two RW-to-address probes; the Q/K/V query projection; the RST query projection; and their DirectTau projections are shared across all L layers. Q and K share read/write rows but retain separate query slices, tau values, applicability coefficients, and outputs. The layer-specific parameters are two LayerNorms and the attention output projection W_ℓ^O . Positional and token embeddings and the final normalization are model-level parameters. Consequently, DAWN-SRW reuses RW primitives across depth without claiming complete block sharing.

3.9. Implemented Addressing and Compute Boundary

For computation type a , define *operator visibility* $\mathcal{V}_a(x)$ as the set of operator identities made available to the control mechanism for applicability evaluation at state x . Visibility denotes the candidate range available to that mechanism, not whether operators are observable to a human analyst. In the evaluated v4172 realization, every operator in the type-specific pool is visible:

$$\mathcal{I}_a(x) \subseteq \mathcal{V}_a(x) = \{1, \dots, N_a\},$$

with $\mathcal{I}_a(x)$ generally a strict subset. The DirectTau boundary determines membership in $\mathcal{I}_a(x)$ within this globally visible pool; it does not change $\mathcal{V}_a(x)$.

The implementation shards operator rows across the model mesh and processes them in dense chunks. The 400M configuration uses one QK chunk, one V chunk, and four RST chunks. Every query scores the full local address pool; every operator row participates in dense read and write contractions before cross-shard reduction. Exact applicability support can be sparse while operator visibility remains global and RW execution remains dense; it is not an indexed sparse kernel. Accordingly, the implemented sequence is global operator visibility, geometrically determined applicability, and dense physical execution.

The evaluated architecture keeps RW execution in the full residual dimension D ; the R -dimensional operation-address space is used only for applicability evaluation. An executor could omit exactly zero RW terms without changing the mathematical output, but such support-restricted RW execution is not implemented or timed here.

4. Experimental Setup

4.1. Models and training

Table 1 summarizes the evaluated checkpoints. The primary DAWN-SRW model has 393,800,708 measured parameters, operation-address dimension $R = 256$, and 48,400 operator rows across QK, V, and RST. It was trained for 39,999,504,384 packed C4 tokens with sequence length 512 and global training batch 1,024. Both dropout rates are zero.

4.2. Matched 400M dense control

The dense control has 393,896,960 measured parameters, width 1,280, feed-forward width 5,120, 18 layers, and 20 heads. It uses the same tokenizer, vocabulary, sequence length, training-token count, seed, batch size, mesh, learning rate, warmup ratio, weight decay, and zero dropout as the primary model. The parameter difference is 96,252, or 0.0244% of the dense model. This comparison tests performance retention under matched scale and training exposure; it is not a formal equivalence test.

4.3. Evaluation protocols

The C4 protocol evaluates 610 complete global batches from the fixed 10M-token capped packed-validation view: 19,520 sequences and 9,974,720 valid next-token targets. Cross-entropy, perplexity, and next-token accuracy use float32 scoring. The zero-shot protocol uses `lm-eval` 0.4.2 with no few-shot examples and no task limit on LAMBADA, HellaSwag, PIQA, ARC-Easy, ARC-Challenge, and WinoGrande. The primary metric is raw accuracy for LAMBADA

and WinoGrande and normalized accuracy for the other four tasks. The mean is an unweighted mean of these six metrics.

4.4. Mechanistic localization and frozen intervention protocol

We define a site by layer, computation type, token position, and global operator identity. IOI discovery uses 119 independent paired-correct units. Sites are ranked by absolute discovery mean contribution importance. We choose the smallest audited prefix that reaches at least 0.70 cumulative absolute importance and 0.95 split top- k overlap. The resulting 4,096-site Q/K/V/RST/layer composition and intervention specification are frozen before validation.

The primary causal score is the summed-log-probability margin between the correct indirect object and the source name. Circuit-wide suppression zeros only the selected operators' execution numerator at target positions while retaining the full production denominator. Four control families each contain 100 frozen-size circuits that exclude circuit sites: equal-count random, same-layer random, activation-matched, and route-frequency-matched across Q/K/V/RST. Exact restoration reinserts the same-example selected numerator after suppression, again under the production denominator. Validation uses 123 independent paired-correct units; the held-out test accessor is opened only after the validation result and unchanged specification are finalized. We report 95% bootstrap confidence intervals and two-sided paired permutation tests where specified.

5. Results

5.1. Language-model performance retention at 400M

Table 2 reports the locked C4 evaluation. DAWN-SRW differs from the dense control by +0.004318918 cross-entropy, +0.081537182 perplexity, and -0.002188432 accuracy (-0.2188 percentage point). These small measured gaps support performance retention, not superiority or formal equivalence.

Table 3 gives task-level zero-shot results with standard errors. The matched 400M means are 0.462872 and 0.459652, a difference of +0.003220495 (+0.3220 percentage point). Task differences vary in sign, so the mean does not establish benchmark superiority. LAMBADA perplexities are 24.0774 ± 0.8005 for DAWN-SRW 400M, 24.0042 ± 0.8073 for dense 400M, and 17.6358 ± 0.5390 for DAWN-SRW 1.3B.

5.2. Descriptive larger-scale evaluation at 1.3B

The 1.272B checkpoint reaches 2.835879 C4 cross-entropy, 17.045382 perplexity, 0.437099 token accuracy, and a 0.485844 six-task mean. It was trained for 19.999B tokens,

Table 1. Model and training specification. The 1.3B row is descriptive and does not form a controlled scaling curve with the 400M rows.

Model	Parameters	Tokens	D	Layers	Heads	QK / V / RST rows	Checkpoint
DAWN-SRW 400M	393,800,708	39,999,504,384	2,304	18	36	4,600 / 14,500 / 29,300	76,293
Dense Transformer 400M	393,896,960	39,999,504,384	1,280	18	20	—	76,293
DAWN-SRW 1.3B	1,272,038,404	19,999,981,568	4,096	24	32	8,534 / 26,904 / 54,352	87,193

Table 2. Packed C4 validation on the identical 9,974,720 target tokens.

Model	CE	PPL	Accuracy
DAWN-SRW 400M	2.940213	18.919873	0.424912
Dense 400M	2.935894	18.838336	0.427100
DAWN-SRW 1.3B	2.835879	17.045382	0.437099

about half the 400M training-token count, and no completed matched dense 1.3B result is admitted to this study. These results therefore describe a larger trained DAWN-SRW model; they are neither a controlled scaling curve nor a dense-model comparison.

5.3. Input-conditioned RW applicability support

At all 9,994,240 packed-C4 positions, Q, K, V, and RST each have strict-subset applicability support at positive applicability margin (Table 4). Mean exact applicability support is 487.896 Q operators, 386.722 K operators, 812.388 in the shared-pool Q/K union, 1,649.994 V operators, and 2,017.703 RST operators. This is exact input-conditioned applicability support under Equation 13.

Applicability support does not make the present kernel efficient. Static forward accounting at batch 1 and sequence length 512 is 4.741260116 TFLOPs for current dense DAWN-SRW, 0.818230250 TFLOPs for dense addressing plus RW application restricted to exact applicability support, and 0.386547057 TFLOPs for the matched dense Transformer. The corresponding opportunity is 17.2577% of current DAWN-SRW accounting but still 2.1168 times the dense Transformer (Figure 2). It is a mathematical conditional-RW opportunity, not measured latency, energy, or hardware speedup.

5.4. IOI circuit localization

Discovery produces split-half rank stability 0.9139788 over 456,697 ranked sites. The smallest prefix passing both frozen discovery gates contains 4,096 sites and 0.719249 cumulative absolute importance. It spans all 18 layers and contains 1,744 Q sites, 2,350 K sites, two V sites, and no RST sites. Split top- k overlap is 0.974854 (Jaccard 0.950941). These statistics identify a Q/K-centered candidate circuit; localization alone is not a causal claim.

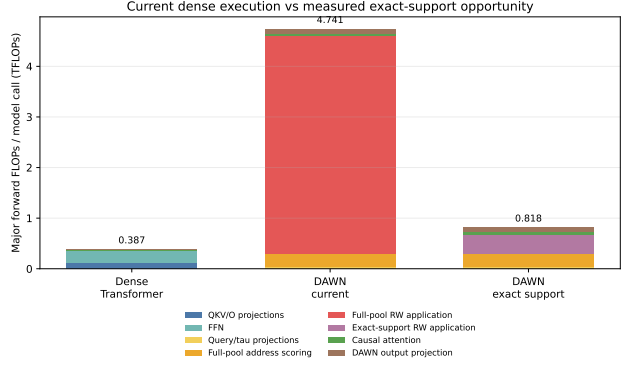


Figure 2. Static backbone accounting at batch 1 and sequence length 512. Address scoring remains dense in the estimate restricted to exact applicability support. Current execution also applies all full-dimensional RW rows. No production latency claim is made.

5.5. Validation and held-out causal intervention

Table 5 shows confirmatory results. On validation, suppression moves the mean margin from 3.465456 to -4.399331 , a drop of 7.864787 (95% CI [7.490080, 8.241263]), and accuracy falls from 1.000000 to 0.008130. The frozen effect exceeds every replicate in all four 100-circuit control families. The most competitive control is activation-matched, with mean drop 5.400490; the frozen-minus-control difference is 2.464297 (95% CI [2.076200, 2.832697], paired permutation $p = 0.00049975$). Exact restoration returns accuracy to 0.991870 and recovers 0.999661 of the suppression effect (95% CI [0.998655, 1.000584]).

The fully held-out test reuses the unchanged site identity, scoring rule, intervention, controls, seeds, and thresholds. Suppression moves margin from 3.254720 to -4.155613 , a drop of 7.410332 (95% CI [7.016485, 7.818472]), while accuracy falls to 0.008772. Again the frozen effect exceeds every replicate in all four control families. The activation-matched mean drop is 5.472579; the frozen-minus-control difference is 1.937754 (95% CI [1.564818, 2.319706], $p = 0.00049975$). Restoration recovers 0.998406 of the effect (95% CI [0.997324, 0.999385]) and restores accuracy to 1.000000.

Together, suppression, matched-control separation, and exact restoration support the claim that this discovery-frozen Q/K-centered operator circuit is causally necessary for the evaluated IOI behavior on independent validation and fully

Table 3. Zero-shot accuracy. Parentheses contain standard errors. HS, AE, and AC denote HellaSwag, ARC-Easy, and ARC-Challenge.

Model	LAMBADA	HS	PIQA	AE	AC	WinoGrande	Mean
DAWN-SRW 400M	0.352028 (0.006654)	0.458375 (0.004972)	0.712187 (0.010563)	0.484007 (0.010255)	0.256826 (0.012767)	0.513812 (0.014047)	0.462872
Dense 400M	0.353192 (0.006659)	0.457080 (0.004971)	0.698041 (0.010712)	0.475589 (0.010248)	0.249147 (0.012639)	0.524862 (0.014035)	0.459652
DAWN-SRW 1.3B	0.395886 (0.006813)	0.510157 (0.004989)	0.723069 (0.010440)	0.509680 (0.010258)	0.268771 (0.012955)	0.507498 (0.014051)	0.485844

Table 4. Exact applicability support at applicability margin > 0 . Q/K union counts an operator in the applicability support for Q or K at the same layer and token.

Computation	Pool	Mean support	Pool fraction
Q	4,600	487.896	10.6064%
K	4,600	386.722	8.4070%
Q/K union	4,600	812.388	17.6606%
V	14,500	1,649.994	11.3793%
RST	29,300	2,017.703	6.8864%

held-out inputs. “Necessary” is intervention-specific: it does not mean the sites form a unique or human-readable algorithm.

5.6. Negative and fail-closed results

The same workflow produces informative negative boundaries (Table 6). RAVEL passes the behavioral gate but fails variable-localization stability, so no circuit is frozen and no confirmatory split is opened. ARC discovery freezes a stable 4,096-site Q/K circuit, but validation suppression has the wrong sign and fails control separation; restoration passes, and test remains closed. For 1.3B IOI, behavioral eligibility and rank stability pass, but the largest allowed 4,096-site prefix reaches only 0.635144 cumulative importance, below the preregistered 0.70 threshold. No circuit is frozen and no validation or test is opened.

6. Discussion

6.1. What the results establish

The constructive test supports three ordered claims. First, the matched 400M comparison establishes viability under the measured protocols. Second, exact input-conditioned applicability support establishes that control is state-dependent in the evaluated model, even though operator visibility remains global and RW execution remains dense. Third, the discovery-frozen IOI result establishes causal accessibility for the specified circuit: suppressing its operator contributions changes behavior, matched controls do not reproduce that effect, and exact same-example restoration recovers it on independent validation and held-out data. RAVEL, ARC, and 1.3B IOI are fail-closed boundaries showing that successful localization is not automatic across tasks or scales.

These results do not establish that language reasoning is a discrete program, that the factorization is intrinsic or optimal, or that identifiable operations have human-readable

semantics. They show that one explicit factorization can support language modeling, exhibit state-conditioned organization, and provide causal intervention targets under the evaluated conditions.

6.2. From the Computational Question to an Open Design Space

If learned high-dimensional computation can be organized as state-conditioned control over operations, what properties should the conditions, operations, their coupling, and their execution have? The present results provide one constructive test in language modeling, not a universal answer. Within this open design space, the narrower architectural hypothesis separates identifiable learned operations from explicit state-conditioned control that determines how those operations contribute to the subsequent state transition. “Identifiable” is an architectural property: an operation is referable as a stable object that can be measured and intervened on. It does not prescribe the scope or duration of that identity and is not a claim of semantic purity. Operations may remain distributed or polysemantic, and a behavior may depend on many operation sites.

v4172 is one explicit realization: its rank-1 granularity, address-based control, cross-depth persistence, and normalized additive composition are design choices rather than requirements of the architectural hypothesis.

What constitutes a learned condition? v4172 represents $C_a(x)$ through an explicit applicability profile derived in a learned operation-address geometry. Other systems could couple applicability directly to functional parameters, use intrinsic read-side activation as in associative-memory-style systems, or use separate control representations. The present evidence does not establish that one organization is preferable.

What constitutes an operation? Rank-1 RW primitives are one granularity. Operations could be finer, coarser, structured, or hierarchical, and their identities could persist over different scopes. Identifiability requires only a referable architectural object available for measurement or intervention; it does not imply semantic purity, and operations may remain distributed or polysemantic.

How should control and operation be coupled? v4172 derives operation addresses from read/write parameters, cou-

Table 5. Frozen IOI circuit results. Accuracy arrows show intact to suppressed. Recovery is the fraction of the suppression effect recovered by exact same-example numerator restoration.

Split	Units	Intact margin	Suppressed margin	Drop (95% CI)	Accuracy	Recovery / restored acc.
Validation	123	3.465456	-4.399331	7.864787 [7.490080, 8.241263]	1.000000 → 0.008130	0.999661 / 0.991870
Held-out test	114	3.254720	-4.155613	7.410332 [7.016485, 7.818472]	1.000000 → 0.008772	0.998406 / 1.000000

Table 6. Fail-closed mechanistic results. Later phases were not opened after a failed preregistered gate.

Study	Passed	Failed boundary	Decision
RAVEL 400M	Behavioral eligibility	Variable stability: 0.628762 / 0.676243 / 0.660622	No frozen circuit; no validation, transfer, or test
ARC 400M	Stable discovery; 4,096 Q/K sites; restoration	Suppression effect wrong sign; control separation failed	Reject causal replication; held-out test closed
IOI 1.3B	Behavioral eligibility; rank stability 0.907188	4,096-site cumulative importance 0.635144 < 0.70	No frozen circuit; validation and test closed

pling functional identity and applicability control without making them identical. Other designs could separate the two more strongly or couple applicability more directly to the transformation parameters.

How should applicable operations compose? v4172 uses simultaneous normalized additive composition. Other computational systems could use sequential, ordered, hierarchical, recurrent, or otherwise structured composition; none of these alternatives is evaluated here.

How much global operator visibility is necessary? The current model exposes the full type-specific operator pool to applicability evaluation. Whether expressive learned control requires global operator visibility or can preserve its conditional structure when operator visibility is candidate-restricted remains an empirical question.

How does logical conditional structure map to physical execution? Applicability support, operator visibility, and physical execution are distinct. The execution implications of this separation are developed below.

6.3. From Logical Conditional Structure to Physical Execution

The present results establish state-conditioned applicability support: for a given representation state, many operators have exactly zero applicability. They do not establish candidate-restricted operator visibility or sparse physical execution. In v4172,

$$\mathcal{I}_a(x) \subseteq \mathcal{V}_a(x) = \{1, \dots, N_a\},$$

so applicability is evaluated over the full type-specific pool before zero-contribution operators are excluded, and RW application remains dense.

The execution problem separates three levels: applicability support, operator visibility, and physical execution. Ap-

plicability support specifies which operations contribute; operator visibility specifies which operator identities the control mechanism can consider when determining applicability; physical execution asks whether the resulting computation can be organized into accelerator-efficient execution units. **Sparse applicability support does not by itself imply hardware-exploitable sparse computation.** Fine-grained state-conditioned control may conflict with hardware-efficient execution granularity, memory locality, dispatch, or communication.

The immediate empirical question is how much operator visibility and executed computation can be reduced, and at what physical execution granularity, while preserving language-model quality and the learned conditional structure. More generally, it remains open whether high-dimensional learned control can retain broad conditional expressivity while acquiring coarse physical locality suitable for efficient execution.

6.4. Limitations

The present results do not establish that transition-producing computation has been factorized into an efficient conditional computational structure. Although each state has strict-subset applicability support, applicability is still evaluated with global operator visibility over large type-specific pools. Performance retention may therefore depend partly on the representational capacity provided by a large distributed operator bank and full-pool applicability evaluation, rather than on a compact or intrinsically efficient decomposition of language-model computation.

More fundamentally, the present results do not establish that condition–operation factorization is a necessary, intrinsic, privileged, or optimal organization of learned high-dimensional computation. They establish only that one explicit realization is viable under the evaluated language-model conditions.

Relatedly, the learned spherical address geometry is not shown to reflect an intrinsic or compressible organization of language operations. It may instead serve as a high-capacity substrate in which useful operators are distributed broadly enough for global state-conditioned access to recover the required transformations. The present study therefore does not establish how much operator visibility is required to preserve language-model expressivity, or whether candidate evaluation and executed computation can be restricted substantially further.

The current implementation evaluates full address and operator pools densely. Exact applicability support does not imply current sparse execution, and even the accounting restricted to exact applicability support remains 2.1168 times the matched dense Transformer’s backbone accounting. No production latency, energy, or hardware advantage is measured.

The central causal evidence is confined to IOI and does not automatically generalize to every task. RAVEL, ARC, and the selected 1.3B IOI replication provide valid negative boundaries. The 400M and 1.3B checkpoints have different parameter and training-token budgets, so they do not define a controlled scaling curve. Operator-level causal units need not map one-to-one to human-readable concepts. Finally, the present work does not establish language reasoning as universally discrete operator programs.

7. Conclusion

This paper begins from a broader computational question: whether computation over learned high-dimensional states can be organized as an explicit, learnable system of state-conditioned control over operations. It tests this question in language modeling through a narrower architectural hypothesis: transition-producing computation can be factorized into identifiable learned operations and explicit state-conditioned control that determines how those operations contribute to the subsequent state transition.

v4172 is one concrete realization, using persistent rank-1 RW banks and a shared geometric rule relating projected state queries to read/write-derived operation addresses. A continuous state-dependent boundary produces exact applicability-support membership, while normalized positive margin determines continuous-valued applicability within the support; the resulting RW contributions compose additively across Q/K/V/RST. The empirical results provide a bounded constructive test: measured 400M viability, descriptive 1.3B evidence, input-conditioned applicability support, and causal accessibility through a discovery-frozen IOI circuit with suppression, matched controls, and exact restoration. Fail-closed results mark where the same localization program did not succeed.

These findings do not establish an intrinsic factorization, human-readable operator semantics, general editability,

benchmark superiority, or current hardware efficiency. They instead motivate an open research program on how learned conditions are represented, how operations are defined and identified, how control and operation should be coupled, how applicable operations compose, how persistence and hierarchy should be structured, and how logical conditional structure can be mapped to efficient physical execution. Candidate-restricted operator visibility and execution are one immediate instance of this broader program.

A. Exact Addressing and Composition Equations

This appendix restates the production computation as a compact reference. Raw read and write parameters are normalized in float32 before being cast for dense contractions:

$$\bar{r}_i = \frac{r_i}{\max(\|r_i\|_2, \epsilon)}, \quad \bar{w}_i = \frac{w_i}{\max(\|w_i\|_2, \epsilon)}.$$

One pair of model-level probes is shared across QK, V, and RST:

$$u_i = \text{unit}(\bar{r}_i^\top P_r), \quad v_i = \text{unit}(\bar{w}_i^\top P_w), \\ k_i = \text{unit}(u_i \odot v_i).$$

The query and threshold use separate direct projections:

$$\hat{q}_a = \text{unit}(xP_a + b_a), \\ \tau_a = -1 + 2\sigma(xu_a + \beta_a), \quad \rho_i = \hat{q}_a^\top k_i.$$

For linear-angular composition:

$$c_i = \text{clip}\left(\frac{\rho_i - \tau_a}{\max(1 - \tau_a, 10^{-4})}, 0, 1\right)$$

and padding rows are masked to zero. This is the single applicability coefficient used in RW composition. Let $M_a = \sum_i c_i$. The production composed output before causal attention or residual addition is

$$F_a(x) = \sqrt{\frac{D}{L}} \frac{\sum_i c_i \langle x, \bar{r}_i \rangle \bar{w}_i}{\max(M_a, 1)^{p_a}},$$

with $p_Q = p_K = 0.5$, $p_V = 1.0$, and $p_{\text{RST}} = 1.2$. Operator rows are sharded over the model axis; local numerators and admission mass are reduced across shards. Chunking changes the contraction schedule, not the equation.

B. Complete Model and Training Settings

The 400M DAWN-SRW checkpoint is the primary performance and mechanistic model. The dense 400M checkpoint is only a matched performance control. The 1.3B DAWN-SRW checkpoint is used for descriptive performance and one fail-closed IOI localization replication. The still-incomplete dense 1.3B run is excluded.

Table 7. Detailed settings from the registered training configurations and restored checkpoint manifests.

Setting	DAWN-SRW 400M	Dense 400M	DAWN-SRW 1.3B
Vocabulary	30,522	30,522	30,522
Residual width	2,304	1,280	4,096
Layers	18	18	24
Heads / head width	36 / 64	20 / 64	32 / 128
Maximum sequence length	512	512	512
Feed-forward width	—	5,120	—
Operation-address width	256	—	256
QK / V / RST pool	4,600 / 14,500 / 29,300	—	8,534 / 26,904 / 54,352
QK / V / RST chunks	1 / 1 / 4	—	1 / 2 / 8
QK / V / RST denominator power	0.5 / 1.0 / 1.2	—	0.5 / 1.0 / 1.2
Initial support-fraction targets	0.05 / 0.03 / 0.01	—	0.05 / 0.03 / 0.01
Training batch	1,024	1,024	448
Mesh (data $\times$ model)	16 $\times$ 2	16 $\times$ 2	16 $\times$ 2
Learning rate	3×10^{-4}	3×10^{-4}	2×10^{-4}
Warmup ratio	0.08	0.08	0.06
Weight decay	0.1	0.1	0.1
Tau learning-rate multiplier	0.001	—	0.001
Dropout / <code>router_dropout</code>	0 / 0	0 / —	0 / 0
Gradient checkpointing	yes	yes	yes
Seed	1	1	1

C. Benchmark Metrics and Protocol Details

C4. All reported C4 rows use the same packed validation artifact and `packed_c4_validation_v1` protocol. The capped view contains 19,531 sequences. Exactly 19,520 sequences form 610 complete global evaluation batches, producing 9,974,720 valid next-token targets; 11 incomplete-tail sequences are excluded before scoring.

Zero-shot. The protocol is `mamba_table3_zero_shot_v1`, using stock `lm-eval==0.4.2`, global batch 32, maximum length 512, length buckets 64/128/256/512, and zero few-shot examples. Evaluated split sizes are 5,153 for LAMBADA, 10,042 for HellaSwag, 1,838 for PIQA, 2,376 for ARC-Easy, 1,172 for ARC-Challenge, and 1,267 for WinoGrande. Task, dataset, tokenizer, and software contracts match exactly between the two 400M models.

D. IOI Localization and Intervention Details

D.1. Frozen discovery

Discovery admits 119 paired-correct independent units and ranks 456,697 sites. Split-half rank stability is 0.913978765. The frozen selection rule is the smallest prefix satisfying cumulative absolute importance at least 0.70 and split top- k overlap at least 0.95. The selected $k = 4096$ prefix has:

- cumulative absolute importance 0.719248713;
- 3,993 shared split sites, overlap 0.974853516, and Jaccard 0.950940700;
- Q/K/V/RST counts = 1744/2350/2/0;
- coverage of every layer from 0 through 17.

Validation and test do not recompute or alter this selection.

D.2. Suppression, controls, and restoration

For a correct candidate c and source candidate s , the primary score is

$$\mathcal{M} = \sum_{t \in c} \log p(t) - \sum_{t \in s} \log p(t).$$

Suppression zeros the frozen sites’ execution numerator at specified token positions. It leaves admission mass and the production denominator unchanged. Controls use the same 4,096-site count and exclude all circuit sites. Activation matching conditions on layer, computation type, and the discovery mean pre-scale operator-output-norm quantile. Restoration executes a distinct intervention: after suppression, it adds back the exact selected same-example numerator under the unchanged denominator.

On validation, restoration returns the margin to 3.462791, leaving residual drop 0.002665. On held-out test, it returns the margin to 3.242905, leaving residual drop 0.011815. The respective recovery fractions are 0.999661155 (95% CI [0.998654544, 1.000584125]) and 0.998405645 (95% CI [0.997323737, 0.999385266]).

E. Negative Mechanistic Results

RAVEL. Discovery uses 142 independent units. Rank stability is 0.628762 for Continent, 0.676243 for Country, and 0.660622 for Language, below the preregistered requirement. No prefix passes the complete discovery rule, no variable circuit is frozen, and validation, transfer, and test are not opened.

ARC. Discovery stability is 0.89386 and a 4,096-site Q/K-only circuit is frozen. On 51 validation units, intact margin is 1.735534 and suppressed margin is 2.819139, giving a

Table 8. Control effects and frozen-minus-control separation. Each control family contains 100 replicates. All paired permutation p -values are 0.0004997501 with 2,000 permutations.

Control	Validation drop	Frozen minus control (95% CI)	Test drop	Frozen minus control (95% CI)
Equal-count	0.070886	7.793902 [7.437851, 8.161572]	0.080276	7.330057 [6.922408, 7.716879]
Same-layer	0.039389	7.825399 [7.440629, 8.180288]	0.041073	7.369260 [6.977447, 7.754539]
Activation-matched	5.400490	2.464297 [2.076200, 2.832697]	5.472579	1.937754 [1.564818, 2.319706]
Route-frequency	0.219162	7.645626 [7.289512, 8.031625]	0.242665	7.167668 [6.779836, 7.545537]

frozen margin drop of -1.083605 (95% CI $[-1.472180, -0.690348]$). The effect has the wrong sign and fails control separation. Restoration recovery is 1.004083 (95% CI $[0.994587, 1.013652]$), but restoration alone does not pass the conjunction. Held-out test remains closed.

1.3B IOI. The scale replication admits 125 independent units and passes rank stability at 0.907187749. The largest allowed 4,096-site prefix reaches only 0.635144 cumulative absolute importance, below the 0.70 gate. No circuit is frozen; validation and test remain closed.

F. Compute Accounting

The applicability-support analysis evaluates 9,994,240 packed-C4 positions and observes address-stage margins before RW application. It stores aggregate statistics, not per-token operator identities or vectors. Static accounting uses batch 1, sequence length 512, and 1 MAC = 2 FLOPs. It counts major matmul/einsum operations and omits softmax, normalization, gate elementwise arithmetic, embedding lookup, residual additions, and communication collectives.

The estimate restricted to exact applicability support is mathematically output-equivalent if the executor can skip all and only zero-weight RW terms. It does not include indexed address retrieval and does not demonstrate kernel parity. Instrumented profiling time is excluded from latency claims because histogram collection is part of the measured path.

G. Reproducibility Identities

The machine-readable checkpoint registry is `configs/paper_checkpoint_registry.yaml`. Exact evaluation commits, software versions, data hashes, result roots, and complete artifact hashes are recorded in the repository experiment ledger. Historical experiment branches are evidence identities only; this source is maintained on the canonical main branch.

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Table 9. Backbone static accounting. “Exact applicability support” retains full dense address scoring and replaces only full-pool RW application with the measured applicability support at positive applicability margin.

Component	Current dense DAWN-SRW	Dense address + exact RW	Dense Transformer
Address scoring + query/tau projections	0.293742	0.293742	–
RW application	4.306187	0.383157	–
Causal attention	0.043487	0.043487	0.024159
Dense output / QKV / FFN maps	0.097845	0.097845	0.362388
Total TFLOPs	4.741260	0.818230	0.386547

Table 10. Primary immutable identities. Branch names are omitted because they move; exact checkpoint steps, hashes, and artifact locations are the binding identities.

Evidence	Exact identity	Repository source
DAWN-SRW 400M	model spatial-r1-v4.1.7.2; step 76,293; checkpoint identity a7ce8afc...f91c17; config hash 08733ae4...5039b	configs/train_config_v4172_400M_c4_40B_v4_64_ver1_den_qk0p5_vlp0_rst1p2.yaml
Dense 400M	model baseline-JAX; step 76,293; config hash c3ce5d72...82fc25	configs/train_config_baseline_tpu_400M_c4_40B_v4_64_tp2.yaml
DAWN-SRW 1.3B	model spatial-r1-v4.1.7.2; step 87,193; checkpoint identity 2e38b41f...bf5f9; config hash 1e877263...4a5222	configs/train_config_v4172_1p3B_c4_20B_v4_64_ver1_den_qk0p5_vlp0_rst1p2.yaml
IOI circuit	parsed frozen-spec hash 6ec26618...418c; site hash 573cfff8...0200; selected-row hash 1439f6f3...fe0c1	configs/paper_ioi_frozen_circuit_v4172_400m.yaml
Compute support	support hash bleac5f5...bdc2; accounting hash d7ad6115...984e	assets/paper/v4172_400m_compute_support_step76293/

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